Arunabha Majumder

Mechanical | Mechatronics Engineer

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EDUCATION

PhD, Mechatronics and Robotics
Aalborg University | 2023 - present
Thesis expected March
Study abroad: NUS Singapore, 2025

MTech, Mechatronics
CSIR-CMERI, India | 2020 - 2022

BTech, Mechanical Engineering
SIT (VTU), India | 2015 - 2019

Diploma in Fine Arts
Bangiya Sangeet Parishad, India

SKILLS

SolidWorks (CSWA) | 3D printing
Stiffness analysis in SolidWorks
Genetic algorithm optimisation
MATLAB/Simulink | LabVIEW | Arduino
Sensor integration | Test rigs
Qualisys motion capture and EMG
Hand sketching | Adobe Illustrator

AWARDS

Best Paper Award, IFToMM ISRM 2026
Best Major Project Award, 2019
TATA Technologies Innovation Grant

PUBLICATIONS

Hybrid actuated shoulder exoskeleton, ISRM 2026 (Springer)
Neural network-based control of a pneumatic muscle, IEEE CONECCT 2022

LANGUAGES

English (C1) | Bengali | Hindi

REFERENCES

Available upon request

PROFILE

Mechanical and mechatronics engineer. I design, build and test robotic hardware and exoskeletons, from the first sketch to a tested prototype.

EXPERIENCE

Research Engineer, Shoulder Exoskeleton (PhD)
Aalborg University, Denmark | *2023 - Present*

- Designed a hybrid exoskeleton (spring + 6 Nm motor) that uses over 25% less energy. Best Paper Award.
- SolidWorks design and genetic algorithm optimisation; stiffness tested on a physical test setup.
- Integrated the motor, encoder and force sensors; built test rigs.
- Designed and modelled a novel 5-bar parallel robot with VSM; estimated its stiffness with less than 10% error and quantified human arm impedance.
- Collected EMG and motion data from 12 people; implemented an Arduino trigger to sync robot and Qualisys.
- Taught university courses and supervised Bachelor's students.

MTech Researcher, Pneumatic Muscles
CSIR-CMERI, India | *2020 - 2022*

- Fabricated and assembled pneumatic muscles and collected test data.
- Designed the outer PID loop of a neural network-based controller in MATLAB/Simulink and implemented it in LabVIEW to drive the muscle (0.3 - 0.78% error).

Bachelor's Project, Rehabilitation Exoskeleton
Siddaganga Institute of Technology, India | *2018 - 2019*

- Designed and built an upper-limb exoskeleton for a paralysed arm: CAD, machining, assembly and testing.